MILITARY STANDARD

TEST EQUIPMENT FOR THE
STANDARD ELECTRONIC MODULES PROGRAM

TO ALL HOLDERS OF MIL-STD-1665:

1. THE FOLLOWING PAGES OF MIL-STD-1665 HAVE BEEN REVISED AND SUPERSEDE THE PAGES LISTED:

NEW PAGE	DATE	SUPERSEDED PAGE	DATE
5	20 March 1980	5 20 March 1980	11 July 1979
6	20 March 1980	6 20 March 1980	11 July 1979
W04-1	20 January 1978	Reprinted without change	

2. The following pages are to be added:

NEW PAGE	DATE
B04-1 and B04-2	20 March 1980
M03-1 and M03-2	20 March 1980
S02-1	20 March 1980
W04-2	20 March 1980

3. RETAIN THIS NOTICE AND INSERT BEFORE TABLE OF CONTENTS.

4. Holders of MIL-STD-1665 will verify that additions indicated above have been entered. This notice page will be retained as a check sheet. This issuance, together with the appended pages is a separate publication. Each notice is to be retained by stocking points until the Military Standard is completely revised or canceled.

Custodians:
Navy - EC
Army - ER
Air Force - 11

Review activities:
Navy - AS, OS, SH
Army - AT, MI, AR
Air Force - 13, 17, 19, 85
DLA - ES

User activities:
Army - AR
Navy - MC
Air Force -

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5963-0004)

FSC 5963

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ITEM 804
ANALYZER, WAVE

TYPE OF EQUIPMENT	Wave analyzer.	
FUNCTION PERFORMED	Frequency selective voltage measurements. Amplitude of tuned signal indicated on analog meter. Tuned frequency indicated by 5 digit LED readout.	

SPECIFICATIONS

Input characteristics		
Impedance	1 MΩ, 30 pF.	
Maximum input level	100 Vrms; ±100 Vdc.	
Amplitude characteristics range		
Linear	30 Vrms to 0.1 μVrms full scale.	
Log	+30 dBV/dBm to -70 dBV/dBm full scale.	
Internal calibration	±1.5%, at full scale, 10 kHz.	
Amplitude accuracy	**LOG**	**LINEAR**
Frequency response 15 Hz to 50 kHz	±0.4 dB	±4%
Switching between bandwidths	±0.5 dB	±5%
Amplitude display	±2 dB	±2%
Input attenuator	±0.3 dB	±3%
Amplitude reference level (IF attenuator)		
Most sensitive range	±1 dB	±10%
All other ranges	±1 dB	±3%
Dynamic range		
Display range	>80 dB.	
Noise sidebands	>70 dB below CW signal 10 bandwidths away from signal.	
Spurious responses	>80 dB for signals less than 0 dBm above 100 Hz.	
Zero response	>30 dB below input reference level.	
IF feedthrough		
Input >10 V	-60 dB or lower.	
Input <10 V	-70 dB or lower.	
Frequency characteristics		
Range	15 Hz to 50 kHz.	
Display accuracy	±3 Hz.	
Typical stability	±10 Hz/hr after 1 hr; ±5 Hz/°C.	
Bandwidths	3, 10, 30 and 100 Hz with a shape factor of 10; 300 Hz with a shape factor of 8.	
AFC pull-in range	>5% bandwidth for 3 Hz through 100 Hz bandwidth; >800 Hz for 300 Hz bandwidth.	
AFC hold-in range	±800 Hz.	
AFC lock frequency	Center of passband ±1 Hz.	
Sweep characteristics		
Frequency span settings	0 Hz, 50 Hz to 50 kHz.	
Sweep time accuracy	±5% of setting.	
Sweep linearity	±1%.	

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Output characteristics (tracking oscillator)	
Frequency response	±3%, 15 Hz to 50 kHz.
Frequency accuracy	±1 Hz relative to center of passband.
Impedance	600Ω.
Level	0 V to >1 Vrms into 600Ω.
THD and spurious content	>40 dB below 1 V signal level.
Operating temperature range	0° to 55°C.
Power requirements	100, 120, 220 or 240 V +5%, -10%; 48 to 66 Hz; 10 watts typical.
Options	Battery supply. Rack mount. X-Y recorder.

EQUIPMENT MEETING ALL SPECIFICATIONS

Manufacturer	Hewlett Packard Model 3581A Wave Analyzer.

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ITEM M03
MULTIFUNCTION METER, ANALOG

TYPE OF EQUIPMENT	Analog multi-function meter.
FUNCTION PERFORMED	Measures dc voltage and current, resistance and charge.
SPECIFICATIONS	
DC volts	0.001 V full scale to 10 V in nine 1X and 3X ranges.
DC current	10^{-14}A full scale to 0.3A in twenty eight 1X and 3X ranges.
DC resistance	100Ω full scale to 10^{13}Ω in twenty three linear 1X and 3X ranges.
Coulombs	10^{-13} coulomb full scale to 10^{-6} coulomb in fifteen 1X and 3X ranges.
Accuracy	
DC volts	±1% of full scale on all ranges exclusive of noise and drift.
DC current	±2% of full scale on 0.3 to 10^{-11}A ranges using the smallest available multiplier setting; ±4% of full scale on $3X10^{-12}$ to 10^{-14}A ranges.
DC resistance	±3% of full scale on 100 to 10^{9}Ω ranges using the largest available multiplier setting; ±5% of full scale on $3X10^{9}$ to 10^{13}Ω ranges.
Coulombs	±5% of full scale on all ranges. Drift due to offset current does not exceed $5X10^{-18}$ coulomb per second.
Meter noise	
DC volts	±25 μV maximum with input shorted on most sensitive range.
DC current	Less than $±3X10^{-15}$A.
Input impedance	Greater than 10^{14}Ω shunted by 20 pF Input resistance may also be selected in decade steps from 10 to 10^{11}Ω.
Offset current	Less than $5X10^{-15}$A.
Input voltage drop	Full scale voltage drop is equal to the multiplier switch setting. Best accuracy is obtained when the smallest multiplier switch setting possible is used.
Multiplier switch ranges	(a) 10 (b) 3 (c) 1 (d) 0.3 (e) 0.1 (f) 0.03 (g) 0.01 (h) 0.003 (i) 0.001

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Polarity- - - - - - - - - - - - - - - - -	Meter switch select left-zero (positive or negative) or center-zero scales. Meter switch does not reverse polarity of outputs.
Battery requirements- - - - - - - - - - -	Six 2N6 (or 246, VS305, NEDA 1602); One RM-1W (1000 hours battery life).

EQUIPMENT MEETING ALL SPECIFICATIONS

Manufacturer- - - - - - - - - - - - - - -	Keithley Model 602.

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ITEM S02
WATTMETER, AC/DC

TYPE OF EQUIPMENT	AC/DC single phase electrodynamometer.
FUNCTION PERFORMED	Measures single phase ac or dc power.
SPECIFICATIONS	
Wattage	
Range	
Series configuration	0.0 to 60 W.
Parallel configuration	0.0 to 120 W.
Accuracy	±0.5 percent (in percentage of full scale).
Frequency	DC and 25 to 500 Hz.
Power factor	20 percent to 100 percent.
Maximum current	
Series fields	1A.
Parallel fields	2A.
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	Singer (Sensitive Research) Model DLW.

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ITEM W04
VOLTMETER, AC

TYPE OF EQUIPMENT	AC voltmeter.
FUNCTION PERFORMED	Broadband ac measurements (indicated rms voltage in proportional to absolute average value of applied sine wave).
SPECIFICATIONS	
Voltage range	1 mV to 300 V in 1, 3, 10 sequence (12 ranges).
Frequency range	10 Hz to 10 MHz.
Accuracy	Over temperature range from 0° to +55°C.
1 mV range at full scale	±1 percent from 40 Hz to 500 kHz; ±2 percent from 20 Hz to 1 MHz; ±4 percent from 2 Hz to 4 MHz; ±4 percent from 10 Hz to 10 MHz; ±10 percent from 10 Hz to 10 MHz.
1 mV range at 1/3 full scale	±3 percent from 40 Hz to 200 kHz; +3,-5 percent from 20 Hz to 200 MHz; +3,-10 percent from 2 to 4 MHz; +4, -10 percent from 10 to 20 Hz.
3 mV to 300 V ranges at full scale	±1 percent from 40 Hz to 1 MHz; ±2 percent from 20 Hz to 4 MHz; ±4 percent from 10 Hz to 10 MHz.
3 mV to 399 V ranges at 1/3 full scale	±3 percent from 40 Hz to 200 kHz; +3, -5 percent from 20 Hz to 4 MHz; +4,-10 percent from 10 Hz to 10 MHz.
Isolation	Instrument may be battery-operated if isolation from power lines is desired.
Input impedance	10 MΩ shunted by 21 pF on 1 mV to 1 V ranges; 10 MΩ shunted by 8 pF on 3 to 300 V ranges.
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	Hewlett-Packard Model 400E*, Model 400 EL.

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EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED

Manufacturer- - - - - - - - - - - - - - - -	Ballantine Laboratories Inc. Model 3046A.
Exceptions - - - - - - - - - - - - - - - -	Function performed - Broadband ac measurements (responds to absolute average value of applied signal), primarily as a dB meter with both dB and voltage scales. Voltage range - 1 mV to 300 V, 100 µV to 300 V with X0.1 range multiplier. 1, 3, 10 sequence (12 ranges), 10 dB between ranges. Frequency range - Usable 2 Hz to 20 MHz, calibrated 5 Hz to 15 MHz. 1 mV range at full scale - ±1% from 40 Hz to 2 MHz; ±2% from 20 Hz; ±(2.0 +2.0) from 10 Hz; ±(1.5+1.5) to 4 MHz; ±(2.5+2.5) to 10 MHz; ±(5+5) from 5 Hz to 15 MHz; ±(% of reading + % of reading), all ranges. Input impedance - 10 MΩ shunted by <24 pF on 100 µV to 300 mV ranges; 10 MΩ shunted by nominally 20 pF on the 1 V to 300 V ranges.

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